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Machine Learning Libraries Tasks

Sales Data Analysis

Use NumPy, Pandas, Matplotlib and Seaborn to complete the following tasks.

Part A: NumPy Tasks

1. Convert Quantity column into a NumPy array.
2. Find Mean, Maximum and Minimum Quantity.
3. Calculate Mean Price and Standard Deviation of Price.
4. Calculate Revenue = Quantity × Price.
5. Find top 3 highest revenues using NumPy.

Part B: Pandas Tasks

6. Load the dataset and inspect it.
7. Display first 5 rows, last 5 rows and dataset information.
8. Check missing values and data types.
9. Create Revenue = (Quantity × Price) - Discount.
10. Find Total Revenue, Average Revenue and Highest Revenue Order.
11. Filter Electronics products, customers age > 30 and Revenue > 20000.
12. Sort dataset by Revenue descending.
13. Group by Category and Region for sales analysis.

Part C: Matplotlib Tasks

14. Bar Chart: Category vs Total Revenue.
15. Pie Chart: Sales Distribution by Region.
16. Line Chart: Revenue Trend by Order Date.
17. Histogram: Customer Age Distribution.
18. Scatter Plot: Age vs Revenue.

Part D: Seaborn Tasks

19. Correlation Heatmap.
20. Count Plot: Orders by Region.
21. Box Plot: Revenue by Category.
22. Pair Plot: Quantity, Price, Revenue and Age.
23. Violin Plot: Revenue by Gender.